

Let ω be a k -form, and let $\vec{v}_i \in V$ be vectors.

$$\text{Alt}(\omega)(\vec{v}_1, \dots, \vec{v}_n) = \frac{1}{k!} \sum_{\sigma \in S_k} \text{sgn}(\sigma) \omega(\vec{v}_{\sigma(1)}, \dots, \vec{v}_{\sigma(n)}).$$

This turns any tensor into an alternating one, and iff ω is already alternating, $\text{Alt}(\omega) = \omega$.

We can define the wedge product of a k -form ω and an l -form η to be

$$\omega \wedge \eta = \frac{(k+l)!}{k!l!} \text{Alt}(\omega \otimes \eta).$$

The factor in front is just a convention so we end up with a determinant. We can calculate that this gives correct results for stuff, for example with $dx \wedge dy$ (where dx and dy are the basis covectors):

$$[dx \wedge dy](\vec{u}, \vec{v}) = 2 \left[\frac{1}{2} dx(\vec{u}) dy(\vec{v}) - \frac{1}{2} dy(\vec{u}) dx(\vec{v}) \right].$$

$$\begin{aligned} & [dx \wedge dy \wedge dz](\vec{u}, \vec{v}, \vec{w}) \\ &= \frac{6}{2} \left[+\frac{1}{6} dx(\vec{u}) [dy(\vec{v}) dz(\vec{w}) - dz(\vec{v}) dy(\vec{w})] \right. \\ & \quad +\frac{1}{6} dx(\vec{v}) [dy(\vec{w}) dz(\vec{u}) - dz(\vec{w}) dy(\vec{u})] \\ & \quad +\frac{1}{6} dx(\vec{w}) [dy(\vec{u}) dz(\vec{v}) - dz(\vec{u}) dy(\vec{v})] \\ & \quad -\frac{1}{6} dx(\vec{w}) [dy(\vec{v}) dz(\vec{u}) - dz(\vec{v}) dy(\vec{u})] \\ & \quad -\frac{1}{6} dx(\vec{u}) [dy(\vec{w}) dz(\vec{v}) - dz(\vec{w}) dy(\vec{v})] \\ & \quad \left. -\frac{1}{6} dx(\vec{v}) [dy(\vec{u}) dz(\vec{w}) - dz(\vec{u}) dy(\vec{w})] \right] \\ &= \det(\vec{u}, \vec{v}, \vec{w}). \end{aligned}$$

This is kinda like expansion by minors? not really.